

UREA CYCLE AND ITS METABOLIC DISORDERS

INTRODUCTION TO UREA CYCLE:

- 1) Since ammonia is toxic to CNS even in traces, liver rapidly removes ammonia from circulation and converts it to a non-toxic water soluble urea. Hence site of urea synthesis is liver.
- 2) The reactions leading to formation of urea from ammonia are proposed by **Krebs and Henseleit**. Hence, urea cycle is also called as **Krebs-Henseleit cycle**.
- 3) Formation of urea from ammonia in urea cycle occurs in five reactions. However the first reaction is not a part of the cycle but for the continuation of the cycle which consist of remaining four reactions product of the first reaction is essential. Further, the intermediates of the four reactions are aminoacids. However no codons exist for them.
- 4) Synthesis of urea from ammonia is a energy dependent process.
- 5) Enzymes of urea cycle are present in mitochondria and cytosol.
- 6) First two reactions of urea formation occur in mitochondria and remaining reactions occur in cytosol.

REACTION SEQUENCE OF UREA FORMATION:

The sequence of reactions involved in the biosynthesis of urea, summarized in five steps as follows:

- 1) Formation of Carbamoyl Phosphate.
- 2) Formation of Citrulline.
- 3) Formation of Argino-succinate
- 4) Formation of Arginine and Fumarate
- 5) Formation of Urea and Ornithine

For the synthesis of urea only one ammonia molecule is used as such. Aspartate serves as donor of another molecule of ammonia. HCO_3^- serves as source of CO_2 required for urea formation.

FORMATION OF CARBAMOYL PHOSPHATE:

- 1) First reaction leading to urea formation is condensation of ammonia and HCO_3^- at the expense of two high-energy bonds to form carbamoyl phosphate. The reaction

is catalyzed by mitochondrial carbamoyl phosphate synthetase-I. The enzyme requires N-acetyl glutamate and Mg^{2+} . N-acetyl glutamate is synthesized from acetyl-CoA and glutamate in the liver. 2ATP molecules are hydrolyzed to 2 ADP and 2Pi in the first reaction. Of the 2Pi one Pi is consumed in the reaction. Since the product of the reaction carbamoyl phosphate is high energy compound its formation thermodynamically pulls subsequent reactions of urea cycle towards urea formation.

REACTIONS OF UREA CYCLE:

- 2) Now the first reaction of urea cycle is catalyzed by ornithine transcarbamoylase. It condenses carbamoyl phosphate and ornithine to **form citrulline**. This enzyme is present in mitochondria. Since the subsequent reactions of urea cycle occurs in cytosol, citrulline formed enters cytosol through specific transporter present in inner mitochondrial membrane.
- 3) Arginino succinate synthetase present in cytosol catalyzes second reaction of urea cycle. It condenses citrulline and aspartate at the expense of two high energy bonds to **form argininosuccinate**. One high energy bond is consumed by the hydrolysis of ATP to AMP and PPi. Further hydrolysis of PPi to 2Pi by pyrophosphatase leads to consumption of another high energy bond.
- 4) In the third reaction of urea cycle argininosuccinate is cleaved by argininosuccinase to **arginine and fumarate**.
- 5) Regeneration of ornithine and formation of urea from arginine is the final reaction of urea cycle. This reaction is catalyzed by arginase.

The ornithine so formed enters mitochondria through specific transporter present in inner mitochondrial membrane to start reactions of urea cycle once again.

Reaction of urea formation is shown in Fig. 12.7.

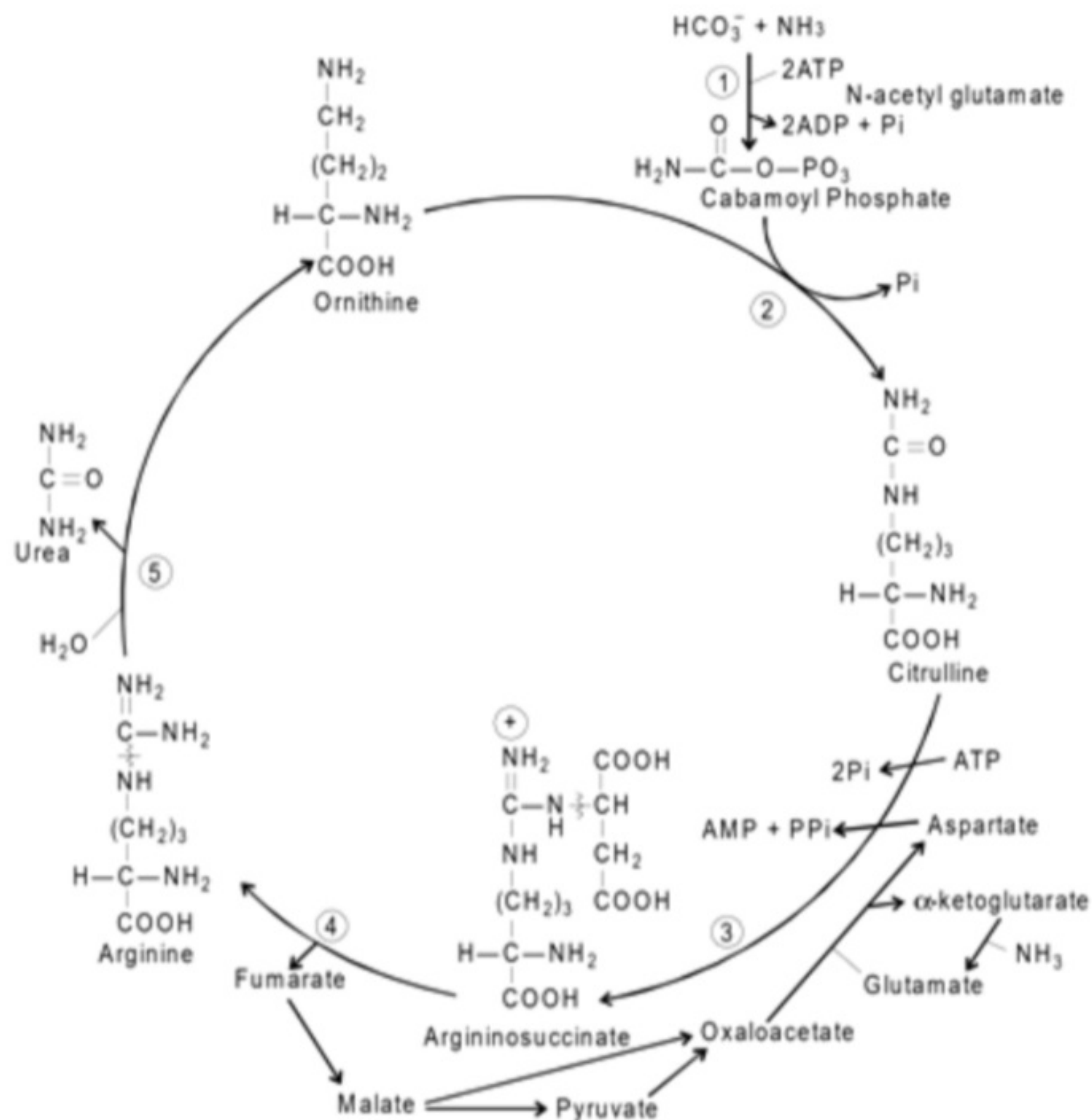


Fig. 12.7 Reactions of urea formation and formation of aspartate from fumarate ~~~~ indicates cleavage

Overall equation for urea formation



FATE OF UREA:

Urea has no physiological function. Hence it is transported to kidneys where it is excreted in urine. It is major end product of protein catabolism in humans. About 10-25 gm of urea is excreted in urine per day which makes up to 80-90% of total nitrogen excreted per day. However, blood also contains some urea.

BLOOD UREA:

Normal blood urea level is 16-36 mg/100 ml.

FATE OF FUMARATE:

Fumarate is recycled in urea cycle by converting it to aspartate. First fumarate is converted to malate by fumarase which is present in cytosol. Malic enzyme generates pyruvate from malate in the next reaction. Next pyruvate is converted to oxaloacetate by pyruvate carboxylase. Oxaloacetate also can be formed directly from malate by the action of malate dehydrogenase. Finally aspartate is generated from oxaloacetate by transamination in which glutamate is converted to α -ketoglutarate. Thus operation of this system cause net supply of one ammonia molecule to urea synthesis (Fig. 12.7).

REGULATION OF UREA FORMATION:

Formation of urea is regulated by activity of carbamoyl phosphate synthetase-I. This enzyme catalyzes committed step in urea synthesis. N-acetylglutamate regulates this enzyme activity. It is an allosteric activator. High protein intake leads to more N-acetylglutamate formation. Thus high protein intake influences urea formation. In starvation also urea synthesis is more mostly due to increased protein breakdown.

METABOLIC DISORDERS OF UREA CYCLE:

Table 14.2: Disorders caused by genetic defects of urea cycle enzymes

<i>Disorders</i>	<i>Defective enzyme</i>	<i>Products accumulated</i>
Hyperammonemia type-I	Carbamoyl phosphate synthase-I	Ammonia
Hyperammonemia type-II	Ornithine transcarbamoylase	Ammonia
Citrullinemia	Arginosuccinate synthase	Citrulline
Argininosuccinic aciduria	Argininosuccinate lyase	Argininosuccinate
Argininemia	Arginase	Arginine

Urea formation is impaired in several inherited diseases. They are due to deficiency of enzymes of urea cycle. The rate of incidence of urea cycle disorders is one in 2500. Most of these inherited diseases are due to defective genes and are fatal. Since the urea cycle converts ammonia to urea these disorders of urea cycle cause ammonia intoxication. Some common clinical symptoms seen in these diseases are vomiting, irritability, lethargy, seizures, mental retardation, coma and early death. They are:

1) HYPER AMMONEMIA TYPE-I:

It is due to deficiency of enzyme carbamoyl phosphate synthetase-I. Mental retardation is the main symptom of this condition.

2) HYPER AMMONEMIA TYPE II:

It is most common among others. It is due to deficiency of enzyme ornithine trans carbamoylase. So, in this condition carbamoyl phosphate accumulates and diverted to pyrimidine formation. This results in excretion of orotic acid and uracil in urine. Glutamate also accumulates in this condition.

3) CITRULLINEMIA:

This condition is due to the absence of enzyme argininosuccinate synthetase. Hence citrulline accumulates in blood and excreted in urine.

4) ARGININO-SUCCINIC-ACIDURIA:

Argininosuccinase is absent in this condition. So, argininosuccinate accumulates in blood and excreted in urine.

5) HYPER ARGININEMIA:

This condition is due to low arginase activity. Hence, arginine accumulates and excreted in urine. However some urea may be excreted in urine due to kidney arginase.

6) N-ACETYL GLUTAMATE SYNTHETASE DEFICIENCY:

It is a rare disorder. N-acetyl glutamate synthetase is involved in formation of N-acetyl glutamate from acetyl-CoA and glutamate. Hyper ammonemia and amino aciduria occurs in this condition. Since carbamoyl glutamate is an analog of acetyl glutamate administration of carbamoyl glutamate can lower symptoms.

TREATMENT:

Treatment of urea cycle disorders involves removal of excess ammonia from blood and reduction of dietary nitrogen load. Peritoneal dialysis is employed to clear ammonia from blood. Administration of compounds which can increase nitrogen excretion is another line of treatment. Benzoic acid and phenyl acetate are two such compounds used in the treatment.